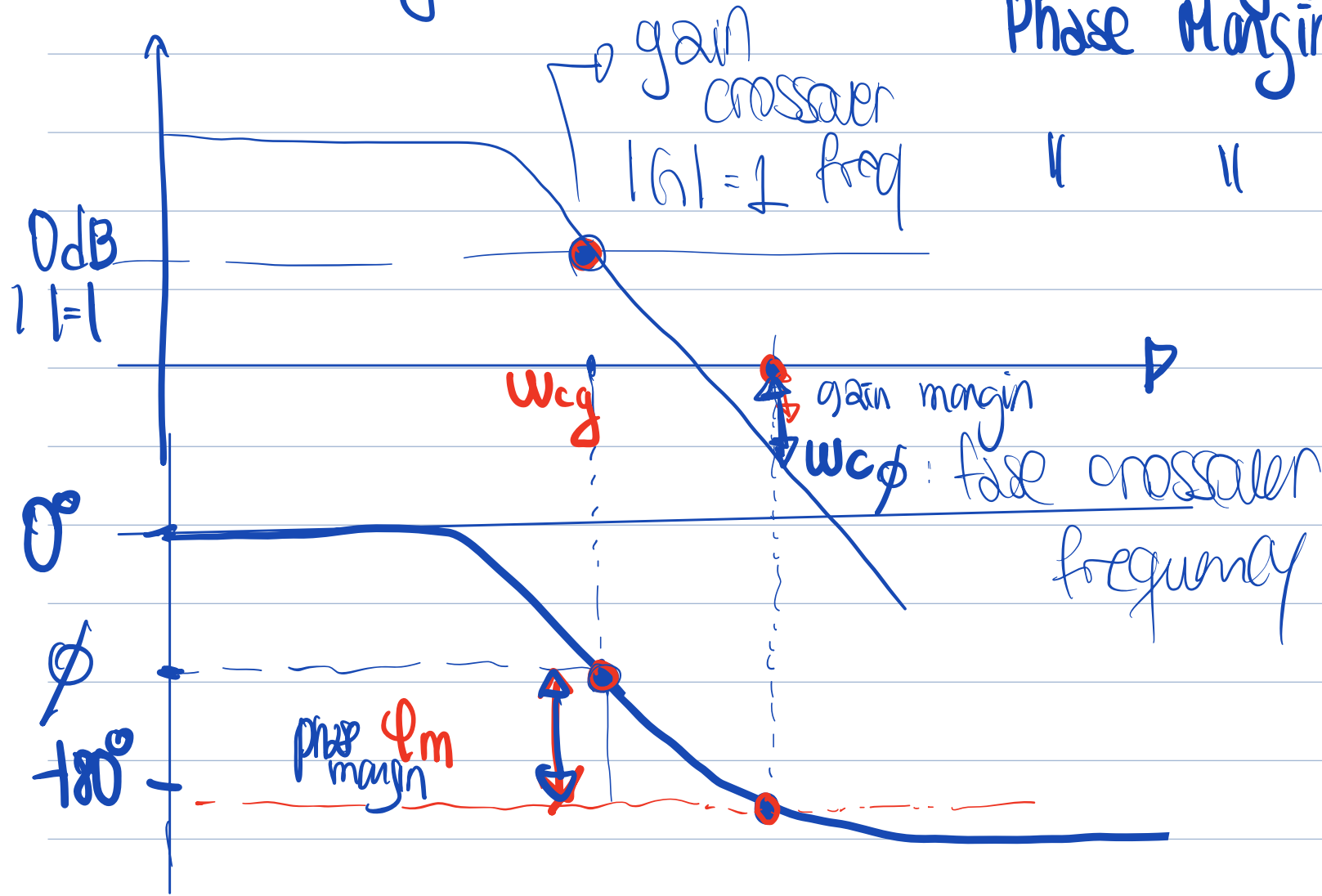
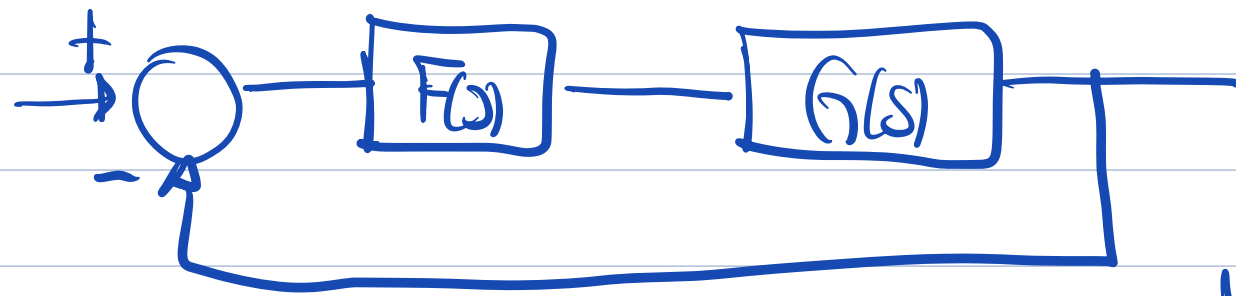


Ex 8 Margin

$\omega_c = \omega_{cg} = \text{gain cross over freq}$
 $\text{Phase Margin} = \phi - (-\pi)$
 $\quad \quad \quad = \phi + 180^\circ$



Compensator Design



We are going to design $F(s)$ in order to:

in order to:

$$\frac{1}{(s + \omega_0)} \Rightarrow \frac{1}{\left(\frac{s}{\omega_0} + 1\right)}$$

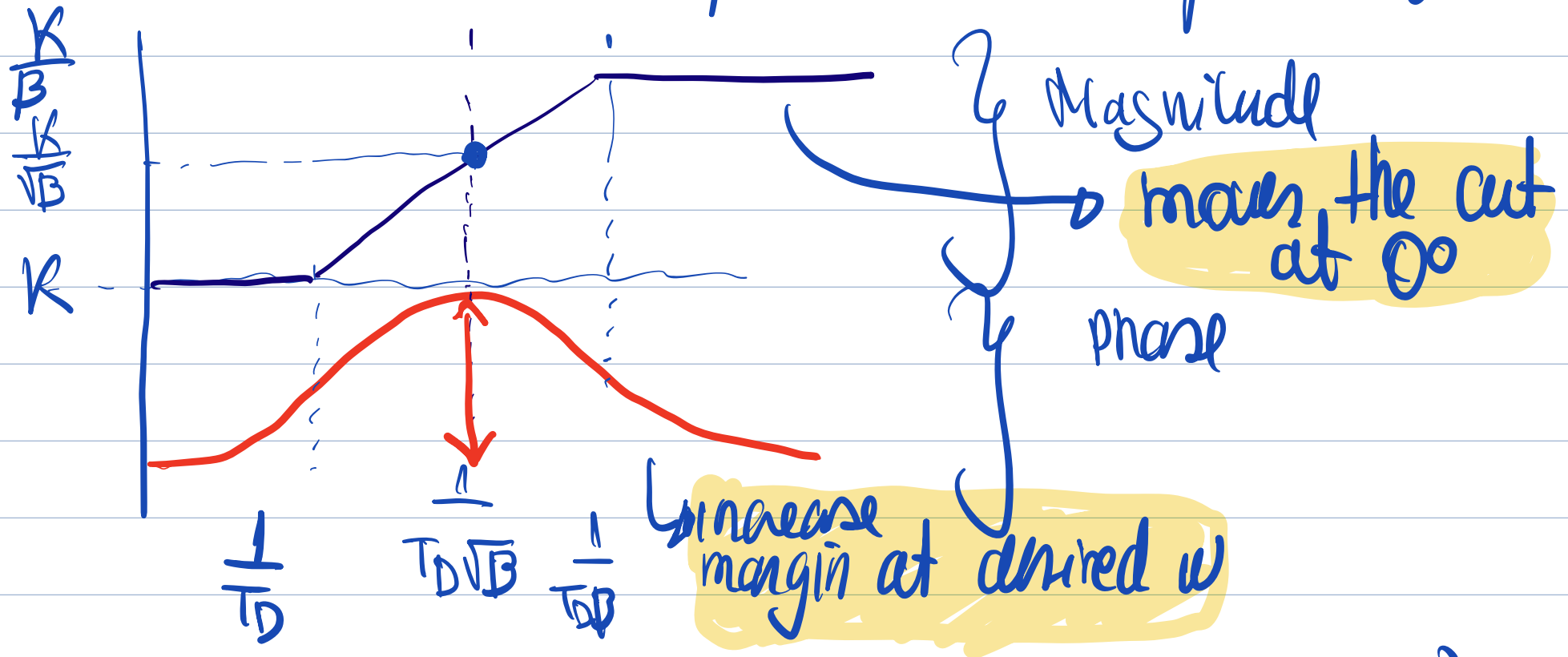
big ω_0 faster system

- Achieve desirable **convergence speed** by means of setting **ω_c** ($\downarrow$ time, $\uparrow \omega_c$)
- Achieve desirable **damping** by setting the phase margin **φ_m** ($\uparrow$ damping, $\uparrow \varphi_m$, $\uparrow$ stable)
- Reduce the **steady-state error** by setting the static gain of the open-loop transfer $|G_o(0)|$

Lead compensator design

$$K \cdot F_{lead}(s) = K \cdot \frac{T_D s + 1}{\beta T_D s + 1}$$

- Fix ω_c (by setting K normally)
- Improves ϕ_m



- β is obtained from the book (plot) of $\Delta\phi$ v/s $\frac{1}{\beta}$

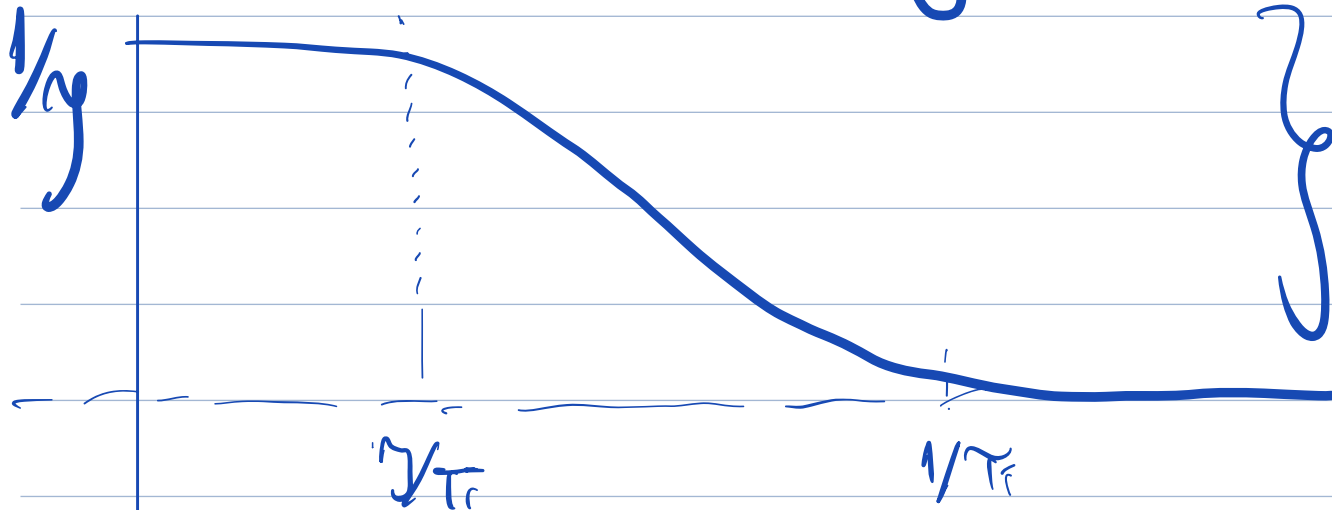
- $\gamma_D = \frac{1}{\sqrt{B} \omega_c} \Rightarrow$ role of thumb!

Lag compensator design

$$Flag(s) = \frac{\tau_i s + 1}{\tau_i s + \gamma}$$

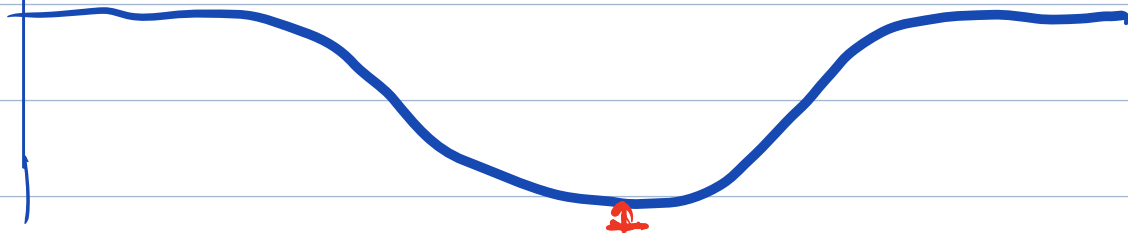
Increase the static Gain $|G(0)|$ of the open loop system, thus reducing the e_{ss}

$$e_{ss} = \frac{1}{1+G(0)} R(s)$$



Determine the phase margin

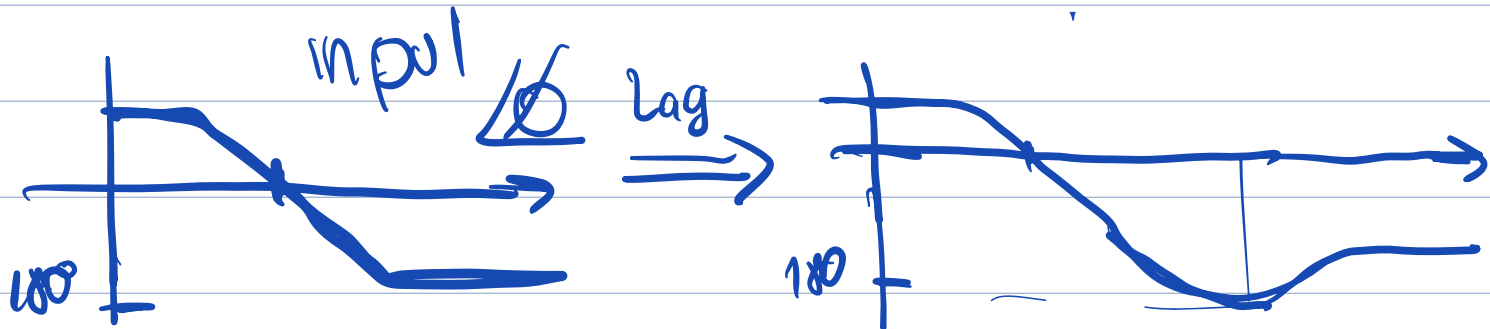
γ



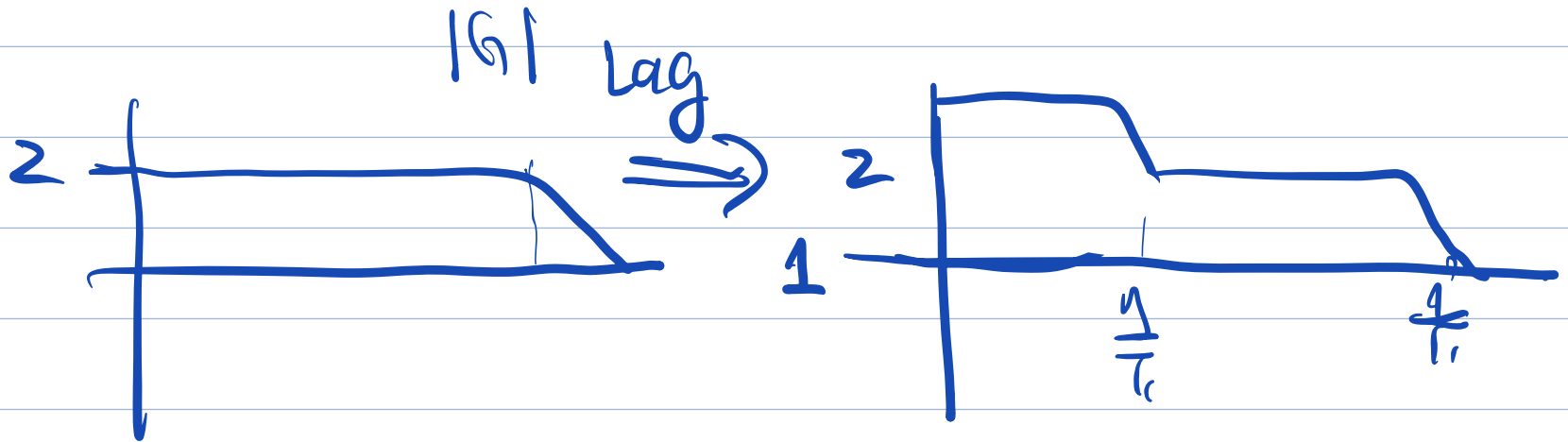
$$\tau_i = \frac{10}{\omega_c}$$

$\Rightarrow \downarrow \tau_i \xrightarrow{\text{open loop}} \Delta \varphi \downarrow \rightarrow \downarrow \text{Stable}$
 $\uparrow \tau_i \rightarrow e_{ss} \downarrow \text{ for small } \omega$

Little τ_i



High τ_i

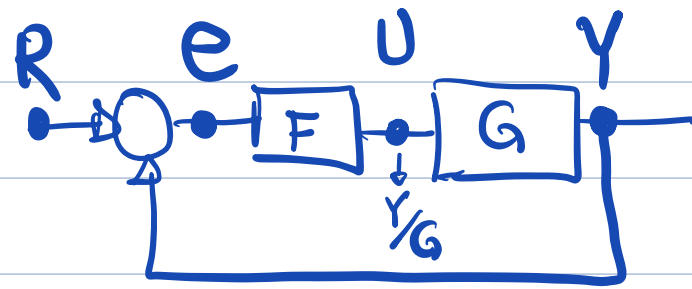


5.20

a)

$$U(s) = F(s) (R(s) - Y(s))$$

$$Y(s) = G(s) \cdot R(s)$$



$$\frac{Y}{G} = F(s) (R(s) - Y(s)) \Rightarrow Y(1 + FG) = R(s) \cdot FG$$

$$\frac{Y}{R} = \frac{F(s) \cdot G(s)}{1 + F(s) \cdot G(s)}$$

b) We have the Bode plot for $G(s)$, therefore for $k=1$, they are asking about stability. Then we can remember that $1+KG$ before stability and we want to avoid -1 for when $k=1 \Rightarrow G=-1$. Therefore, looking at the plot for 0dB or $|G(s)|=1$ the angle is -130°

$$\varphi \Delta = \varphi - (-180) = -130 + 180 = 50^\circ \quad \text{It's stable}$$

c) closed loop twice as fast
 From graph we know that for $K=1$ and 0 dB is in $\omega=1$, then we need those 0 dB to be in $\omega=2$. (we need to make the crossover frequency that is 2 times bigger)
 When $\omega=2 \Rightarrow |G(j\omega)| = 0.3$, therefore
 to scale it to 1 or 0 dB we just
 need to take $K = \frac{1}{0.3} = 3.33$

d) when we make it faster we get an
 angle of -175° , then $\Delta\varphi = -175 + 180 = 5^\circ$
 we are close to instability, ... we lose stability

$$e) F_{lead}(s) = K \cdot \frac{T_D s + 1}{T_D \beta s + 1} \Rightarrow K \cdot \frac{\gamma_D s + 1}{T_D \beta s + 1}$$

closed-loop system $\times 2$ faster, same overshoot as with $K=1$... overshoot related with the m phase! (remember complex poles)
Then, $\Delta\varphi_{K=1} = 50^\circ$ but we have $\Delta\varphi_{w=2} = 5^\circ$

- we need to increase it to $45^\circ \Rightarrow$ By the graph, the maximum phase increase is given by $\frac{1}{\alpha} = 5.55 \Rightarrow \beta = 0.18$ with $K = 3.33$

- T_D is given by $\frac{1}{\sqrt{\beta} \omega_c} = \frac{1}{2 \cdot \sqrt{0.18}} = 1.18$

we have to make sure that the $G(s)$ is $\downarrow$ given $w=2$ since we construct it using

the Gol Bode, then

$$|F(j\omega) \cdot G(j\omega)| = 1 \quad / \quad \omega = 2$$

$$K \cdot \left| \frac{T_D j\omega + 1}{T_D j\omega + 1} \cdot G(\omega = 2) \right| = 1$$

$\downarrow$ 0.3

$$K = 1.4$$

$$f) \quad E(s) = \frac{1}{1+GF} R(s)$$

$$\lim_{t \rightarrow \infty} e(t) = s E(s) = \frac{s}{s \rightarrow 0} \frac{1}{1+GF} = \frac{1}{1+G(0) \cdot F(0)}$$
$$\frac{1}{1+k \cdot G(0)} = \frac{1}{1+1.4 \cdot 3.3} = 0.178$$

If we want to reduce the e_{ss}

$$\frac{1}{1 + K \cdot 3.3} < 0.178 \Rightarrow K > 1.4$$

$$g) \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s \cdot E(s) = \cancel{R(s)} s \cdot \frac{1}{1 + F(s) \cdot F_{lead}(s)} \Big|_{s=0}$$

$$s=0 \quad = \frac{1}{1 + 1.4 \cdot \underbrace{F(0)} \cdot 3.3} = 0$$

the only way to get $e_{ss} = 0$ is making $F(s) \rightarrow \infty$

$$\frac{1}{1 + F(s)} \quad F(0) \Rightarrow \infty \Rightarrow K \cdot \frac{1}{\gamma} = \infty$$

$$\frac{1}{1 + F(s)} \approx 0$$

$$\Rightarrow \gamma = 0 //$$

n) Now we are given $e_{ss} = 0.01$

$$\frac{1}{1 + 1.4 F(0) \cdot 3.3} = 0.01$$

$$100 = 1 + 1.4 \cdot F(0) \cdot 3.3$$

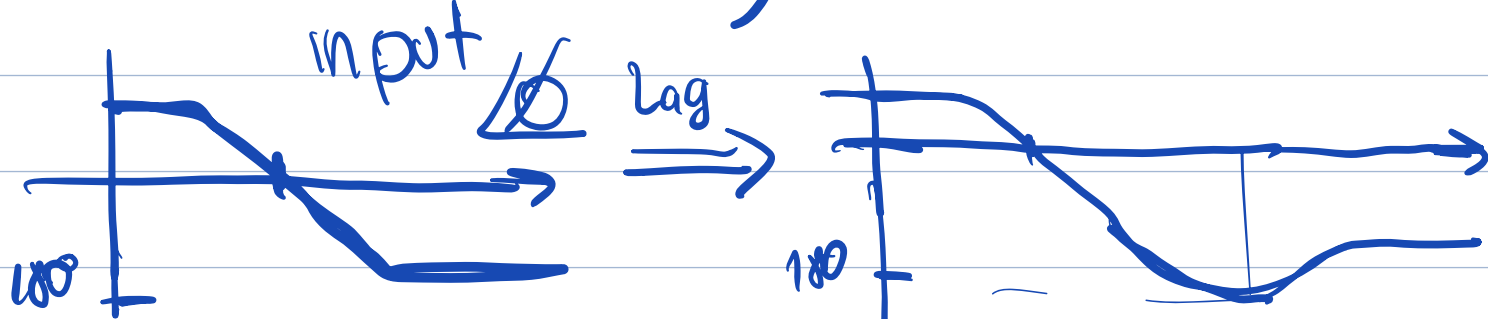
$$F(0) = 21.42 = \frac{1}{\gamma} \Rightarrow \gamma = 0.047$$

$$\tau_i = \frac{10}{2} = 5$$

If $\uparrow \tau_i$
margin
closed

cross freq. increase and phase
will decrease more (more
than 180°)

Little τ_i



High γ_c

